

Research cutoff: September 8, 2026. All web sources retrieved on or before this date; several "current" pages reflect 2026 site content, which is noted where relevant.

1. Feature ideas with evidence

No dedicated academic literature isolates individual FBS tempo/efficiency features against sportsbook totals with documented out-of-sample results. Most cited evidence below is either mechanism-only (explains scoring), raw-score evidence (correlates with points scored), or NFL/soccer transfer evidence. Direct FBS totals-beating evidence is essentially absent for feature-level claims; this gap is stated explicitly per feature group rather than inferred.

Tempo and possession features

Plays per game, seconds per play, and expected possessions measure how many scoring opportunities a game generates; more possessions mechanically raise the ceiling on total points, holding efficiency constant. Construction: trailing N-game or season-to-date plays/possessions per 60 minutes, adjusted for opponent pace and game state, computed only through the prior completed game (never including the game being predicted). Lookback windows of 4-8 games balance recency against noise; exponential/recency weighting (e.g., half-life of 3-4 games) is a reasonable but empirically untested choice for CFB [uncertain]. Opponent-adjustment requires isolating a team's "neutral" pace from opponent-forced pace (e.g., regression of plays-per-game on opponent's typical plays allowed). Shrinkage toward conference or FBS-wide mean is advisable for teams with under ~4 games of current-season data.

The 2023 NCAA clock rule changes (running clock after first downs outside the final 2 minutes, no consecutive team timeouts, shortened overtime procedures in earlier years) mechanically reduced plays per game by roughly 7.8% and shortened games by about 1.4% in an analysis of the first weeks post-implementation. This is a documented mechanism/raw-scoring-adjacent effect (fewer plays should mechanically reduce expected total points, all else equal) but it is not direct totals-market evidence; whether the market fully priced this shift by the time it activated is untested here [uncertain]. Because this is a structural regime break, pre-2023 and post-2023 pace data are not statistically interchangeable and must be flagged in any longitudinal model.[^1][^2][^3][^4]

Garbage-time distortion is a documented issue for how ratings systems compute pace and efficiency: analysts explicitly filter for "garbage time" (blowout, late-game, running-clock situations) when building predictive ratings, because scoring behavior and pace change once a game is decided. One forum-level tally found roughly 16% of total points in a data set came in garbage time by one definition [anecdotal] (undocumented methodology, single-source tally). This is mechanism-level evidence that garbage-time-inclusive average pace/efficiency numbers are noisy proxies for true team ability, which matters for feature construction but says nothing about market inefficiency.[^5][^6][^7]

Interaction of both teams' pace (offense pace vs. opponent's typical allowed pace) is a natural multiplicative/additive feature for expected plays in the specific matchup; this is mechanism-only, untested at the CFB totals level here.

Offensive and defensive efficiency (EPA, success rate, explosiveness, havoc, turnovers, finishing drives, red zone, special teams)

EPA per play and success rate are the standard advanced-efficiency currencies in modern CFB analytics writing, explicitly built to estimate the point value of play outcomes relative to down/distance/field position expectation — this is a raw-scoring-relevant, mechanism-grounded construction, not direct totals-market evidence. Community discussion states success rate captures playing consistently while EPA also captures explosive-play value, and that EPA has strong predictive relationships with scoring [anecdotal], since no peer-reviewed CFB study was found quantifying EPA's marginal predictive power over the market total specifically. No direct FBS totals-market evidence exists for EPA, success rate, explosiveness (IsoPPP-style), havoc rate, turnover rate, red-zone efficiency, fourth-down aggressiveness, or special-teams scoring as standalone total-beating signals; all are level-1/level-2 evidence (mechanism/raw-score) only. Turnover rate specifically should be treated with caution because in-season turnover rates are known to regress heavily toward the mean (a widely cited but not directly-sourced-here regression phenomenon) — treat turnover-driven scoring swings as largely luck-driven and heavily shrink year-over-year and even within-season estimates.[^8][^9][^10]

Construction for all efficiency metrics: entering-game trailing average (season-to-date or trailing 4-8 games) with opponent adjustment (e.g., regress raw efficiency against opponent's average efficiency allowed, iterating to convergence similar to a Massey/SRS-style adjustment), and shrinkage toward FBS or conference-tier mean scaled by sample size (games played). These metrics generally have moderate leakage risk if computed from box scores that are finalized only after each game (which is standard) but severe risk if season-long or full-season "final" ratings are used as inputs to games earlier in that same season.

Matchup interaction features

Pass-offense-vs-pass-defense, rush-offense-vs-rush-defense, pace-vs-pace, and explosive-offense-vs-explosive-defense-prevention are standard multiplicative or differenced interaction terms built from the efficiency features above; no dedicated FBS totals study was found testing these interactions directly against market total, so this is mechanism-only. Team-strength asymmetry (spread magnitude) interacting with total is empirically well-documented in market-structure literature discussed in Section 2 (favorite-side bias, censoring bias), which is a stronger evidentiary basis than any raw interaction feature discussed here.[^11][^12][^13]

Weather, venue, geography, and environment

Historical observed weather cannot be used as a pregame feature unless it is replaced with a timestamped archived forecast matching the intended decision time; this is a hard requirement stated by the user and confirmed necessary because weather forecasts and actuals routinely diverge, especially at longer lead times. Visual Crossing explicitly offers a "Historical Forecast" product — weather forecasts as they existed on past dates — as a separate, paid add-on tier (\$165/month cited on its pricing page), distinct from its free/cheap historical-observed-weather tier (1,000 records/day free, 30,000/month). This confirms that archived forecast data exists but is not free; observed historical weather is comparatively cheap or free via Meteostat and Visual Crossing's non-forecast historical tier. Because archived forecasts cost materially more, a solo developer should expect either a paid subscription for retrospective forecast-consistent backtesting or, more realistically, to run weather features only prospectively from the point of adoption forward, using observed-weather retrospective analysis only to establish plausible mechanism magnitude, not to build a legitimate opener-time backtest.[^14][^15][^16][^17][^18][^19]

Wind is the most commonly cited weather variable affecting scoring. Industry analysis states that games with winds of 13 mph or higher have hit the under at roughly 57% across a large sample since 2005 in one outlet's dataset, and a separate outlet reports the under going 971-816 (54.3%) in all games with wind over 15 mph, rising to 258-193 (57.2%) with an 11.3% cited return when narrowing further. Both of these are [anecdotal]: no disclosed methodology, no confirmed line-timing (opener vs. close), no confirmed out-of-sample/holdout structure, no adjustment for vig, and likely subject to threshold-mining (the narrower subgroup with better win rate is the kind of post-hoc cut this report's methodology explicitly warns against). They should be treated as directionally consistent with a plausible mechanism (wind disrupts

passing efficiency and kicking) but not as validated betting evidence. A qualitative discussion of NFL weather effects states that wind affects passing production more than passing efficiency and that scoring as a whole should decrease as wind increases, while noting most weather conditions have only "light to moderate" effects overall — this is an NFL, not CFB, source, so it is a transfer hypothesis, not confirmed CFB effect, and is largely qualitative rather than quantified with a defensible threshold. No credible source in this research located a rigorously validated wind, temperature, or precipitation threshold specific to FBS with disclosed sample size, out-of-sample testing, and vig-adjusted returns; all specific numeric wind thresholds found here are [anecdotal] and should be labeled [uncertain] for actual predictive magnitude.[^20][^21][^22]

Dome/retractable roof status, elevation, travel distance/time-zone change, and neutral-site status are mechanism-plausible (domes remove wind/precipitation variance; altitude affects ball flight and fatigue; travel and circadian mismatch may affect execution) but no FBS totals-specific quantitative study was found in this research; these remain mechanism-only or, at best, anecdotal industry commentary.

Schedule and roster features

Returning production (ESPN/Bill Connelly methodology) is a widely used preseason-strength proxy that explicitly measures the percentage of prior-year production (yardage/scoring/tackles, etc., weighted by position) returning to a roster, and ESPN's 2026 leaderboard highlights that returning production is at an all-time low industry-wide due to transfer portal and early player movement. CBS Sports separately tracks "returning snap percentage" as a related but distinct, non-position-weighted continuity metric. These metrics are published preseason (before Week 1) and are therefore available at the correct decision time for early-season games, but they are constructed retrospectively from the prior completed season and therefore carry a real revision/leakage risk if the underlying source updates its methodology or data after publication; developers should archive the exact preseason snapshot rather than re-scraping later in-season.[^23][^24]

Recruiting/talent ratings show a documented statistical relationship with team strength: one cited analysis found recruiting-class ratings explained up to 36% of variation in final-season Sagarin ratings in a longitudinal study, and a separate piece notes recruiting rankings are useful predictors of subsequent-season performance [anecdotal cited-source, not fully verified original methodology]. This is raw-scoring/team-strength evidence, not direct totals-market evidence, and recruiting ratings are, at best, one input among many correlated with market-embedded team quality (the market total already substantially reflects team strength via consensus perception), so recruiting data likely proxies information already in the market line more than it adds new information — flag as "may proxy market" rather than "adds new information," pending direct testing.[^25][^26]

Transfer portal and roster turnover: one social-media-sourced claim states roughly 30-38% of an FBS roster changes annually via transfers, graduation, and NFL departures [anecdotal, unverified primary source]. Academic and blog sources note eased transfer rules have "drastically impacted" player movement, and one university blog analysis states that for non-Power-conference programs, increasing roster transfer percentage is associated with increased wins on average, while effects differ by conference tier [uncertain – methodology and significance testing not independently confirmed in this research]. A new "Brown Talent Index" model attempts to combine recruiting pedigree, prior production, and projected playing time into a single roster-talent percentile, illustrating an emerging but unvalidated alternative to recruiting ratings alone; no totals-market backtest was found for this or any competing talent index.[^27][^28][^29][^30][^31]

Injuries, depth charts, and QB starter status: no free, comprehensive, timestamped historical archive of FBS injury reports or depth charts covering 2012-2025 was identified in this research. Unlike the NFL (which has official injury reports), FBS injury reporting is inconsistent by school and conference, frequently informal (beat-reporter tweets, press conferences), and not centrally archived with timestamps in a research-grade format. This is a genuine data gap: state explicitly that historical as-of injury/depth-chart data for FBS is largely unusable for retrospective research at scale, and any injury feature is realistically a prospective-collection-only project going forward (see Section 5).

Market-derived features

Opening total, current total, and price/juice are, per the user's own audit requirement, fields whose provenance (specific sportsbook, opening vs. closing designation, exact timestamp) must be verified directly against the CFBD schema before use; this report does not have direct confirmation of CFBD's line-field metadata beyond the general API documentation reviewed, which describes tiered access and historical coverage claims but does not, on the pages reviewed, provide field-level documentation of sportsbook identity or timestamp granularity for the lines endpoint. This is flagged as unresolved and requires direct empirical inspection of the stored CFBD line records (recommended first diagnostic step, Section 6) before any market-derived feature is trusted.[^32][^33]

Historical FBS odds with sportsbook identity and timing are available via The Odds API's historical endpoint, but explicitly only from mid-2020 onward for NCAAAF featured markets (moneyline, spreads, totals), which is a much shorter historical window than the user's 2012-2025 CFBD dataset — meaning the CFBD-stored line fields are the only plausible source for pre-2020 lines, and their exact sportsbook/timestamp fidelity must be independently audited rather than assumed.[^34][^35][^36][^37]

Feature-group priority table

Feature group	Potential mechanism	Direct FBS totals evidence	Historical availability	Opener-time availability	Leakage risk	Build effort	Priority
Tempo/possessions	More plays → more scoring chances	None found	CFBD play-by-play 2012-2025 (coverage varies by year)	Yes, trailing-average construction	Low if strictly trailing	Medium	High
EPA/success rate/efficiency	Direct point-value estimate of play outcomes	None found; mechanism-grounded[^9][^10]	CFBD-derived, computable from PBP	Yes	Medium (must exclude garbage time carefully)	Medium-high	High
Garbage-time adjustment	Removes non-representative scoring/pace	Mechanism-only[^5][^6][^7]	Computable from PBP + score margin/clock	Yes	Low	Medium	Medium-high
2023 clock-rule regime flag	Structural pace/scoring shift	Raw-scoring evidence, non-market[^2][^3]	Known date	Yes	Low	Low	High (as control variable)

Feature group	Potential mechanism	Direct FBS totals evidence	Historical availability	Opener-time availability	Leakage risk	Build effort	Priority
Weather (wind, temp)	Passing/kicking disruption	Anecdotal thresholds only ^{[^21][^22]} ; NFL qualitative ^[^20]	Observed: cheap/free ^[^14] ^[^15] ; forecast-archive: paid ^[^16]	Only if forecast-timestamped source purchased	High if observed weather misused as forecast	Medium (paid data)	Medium
Returning production / recruiting talent	Preseason team-strength proxy	Raw-strength correlation only ^{[^23][^25]}	Preseason published annually	Yes, if archived at publication	Medium (methodology revisions)	Low-medium	Medium
Transfer portal / roster turnover	Personnel-quality shock	Weak, unverified ^[^30] ^[^31]	Emerging trackers only, short history	Partial	High (new phenomenon, thin history)	Medium-high	Low-medium
Injuries/depth charts/QB status	Talent availability	No usable historical archive found	Not available historically at scale	Prospective only	Very high retrospectively	High	Low (retro), Medium (prospective)
Market total/spread/price	Consensus forecast	Strong indirect (Section 2)	CFBD claims 2012-2025; field fidelity unverified	Depends on audited timestamp	High if close mixed into opener features	Low-medium	Critical (must audit first)

2. Market-structure findings for FBS totals

Documented academic findings

Paul and Weinbach (2005) find that betting markets for college football (and arena football) totals show evidence of "bettor preference for scoring," implying the market skews total prices to accommodate public appetite for overs rather than pure unbiased forecasting. A related summary states the authors "conclude betting the over is systematically overpriced," i.e., the over is a poor long-run bet in the total markets they studied. This is characterized in the secondary source as raw market-bias evidence for college football team-totals betting, not a spread/total distinction confirmed independently here; the primary abstract is behind a paywall, so this report relies on secondary description [\[limitation: abstract/summary only\]](#).^{[^38][^11]}

Arcott (2023, Journal of Sports Economics) is the most directly relevant peer-reviewed CFB totals paper found. It shows that gambling-market score predictions embedded in team-totals and spread lines are biased due to "censoring at zero points" (scores cannot go negative, which distorts the market's effective probability distribution near the boundary), and that this bias creates arbitrage-style opportunities on the team-totals line specifically. The paper reports a naive strategy exploiting this bias nets "over 55 percent" win rate over roughly two decades of data, which the authors state "significantly exceeds the typical transaction costs" (i.e., beats vig). This is Level 4 evidence (positive expected value after transaction-cost consideration) but specifically for the team-totals market (individual team score over/under), not confirmed identically for the combined game total; it also is an older working paper/SSRN version cited alongside the 2023 published version, so replication status across the two versions should be checked directly before relying on exact figures. Season count, exact bookmaker, and line-timing (opener vs. close) are not confirmed from the snippets retrieved — "Not reported in source" pending full-text review. This is genuine peer-reviewed evidence of CFB market inefficiency, but it is about team totals, which the user's own scope explicitly allows as an intermediate construction — it does not directly establish inefficiency in the standard combined-game total line that sportsbooks post as "the total."^{[^39][^12][^13]}

An earlier CFB-specific market efficiency study (Coleman, Lyon, or similar AEA working paper) tested spread market efficiency using 11,000+ games from 1985-2003 and found systematic favorite overpricing. This is spread-market evidence, explicitly excluded from being treated as totals evidence per this report's rules, and is only referenced here to show favorite/underdog bias exists in FBS pricing broadly, as background context for market-maturity claims — it must not be conflated with totals mispricing.^[^40]

NFL totals literature (transfer hypothesis only)

Shank (2018, Journal of Economics and Finance) finds statistically significant inefficiency in both the NFL spread and totals markets, including a documented pattern that games are more likely to hit the over when the closing total is unusually small or unusually large (a quadratic/U-shaped relationship to game outcome), and that a team is more likely to hit the over again if it hit the over in its prior games (a momentum-style bias). Paul and Weinbach (2002) find that under bets on NFL totals win more than 50% of the time historically (1979-2000) but that a statistically "fair bet" (efficiency) cannot be formally rejected once standard error is considered, i.e., the edge is directionally suggestive but not conclusively proven profitable. Kelly and West's NFL totals study similarly finds the NFL totals market is specifically inefficient at the highest posted totals, with bettors showing a preference for betting the over in those high-total games. A separate preseason-specific NFL totals study covering 1996-2019 instead finds market efficiency overall, with the under covering only 50.8% of preseason games (statistically indistinguishable from a fair coin after vig) — this is an important null/conflicting result showing that "totals markets are inefficient" is not a universal finding even within the NFL literature, and it should be weighed against the positive findings above rather than cherry-picked out.^{[^41][^42][^43][^44][^45]}

All NFL findings above are explicitly transfer hypotheses for CFB, not confirmed CFB effects, per this report's rules. The directional consistency (over-preference bias, high-total mispricing, extreme-total bias) across several independent NFL papers strengthens plausibility that similar dynamics could exist in CFB, but none of it should be treated as direct evidence.

CLV concept and mechanics

Closing line value literature (largely industry/practitioner sourcing, not academic) describes CLV as the gap between the bettor's line/price and the eventual closing line/price, typically regarded in the sports-betting industry as the most reliable indicator of long-run skill because it isolates forecasting quality from short-run variance. This entire body of CLV explanation is [\[anecdotal\]](#) industry practitioner content, not peer-reviewed, and should be understood as a widely-repeated heuristic rather than a rigorously proven causal link between CLV and long-run realized profit — the report's own evidence rules require this distinction be preserved.^{[^46][^47]}

Wind/weather and public-narrative claims

The wind-related under-hitting statistics cited in Section 1 (57% under at 13+ mph since 2005; 54.3%-57.2% depending on threshold since some baseline) reappear here as market-bias claims rather than pure scoring claims: if true and executable, they would imply the market under-adjusts totals for wind. Both sources are [\[anecdotal1\]](#): no disclosed sportsbook/line-timing, no disclosed out-of-sample structure, unclear whether thresholds were predeclared or mined after seeing results, and no adjustment for vig in the return figures shown. They should not be treated as demonstrated CLV or profit evidence.^{[^21][^22]}

Market-segment evidence table

Market segment or claim	Evidence type	Line timing	Sport/seasons	Result type	Vig/execution treatment	Evidence quality	Practical implication
CFB team-totals censoring bias	Peer-reviewed paper	Not reported in source	CFB, ~20 years to 2022/2023	Win rate >55%, beats transaction costs	Costs addressed qualitatively ^[^39]	High (peer-reviewed) but scope = team totals, not game total	Worth direct replication on CFB data before trusting
CFB bettor over-preference / overpriced over	Peer-reviewed paper (secondary description)	Not reported	CFB, arena football, ~2005 study period	Directional bias	Not reported	Medium (abstract-level access only)	Treat over-side skepticism as a prior, not a rule
NFL high/low total quadratic bias	Peer-reviewed paper	Not reported	NFL, study through ~2018	Statistical inefficiency	Reported as statistically significant ^[^41]	Medium-high, NFL only	Transfer hypothesis for CFB extreme totals
NFL preseason totals efficiency	Peer-reviewed/working paper	Not reported	NFL preseason, 1996-2019	Null result (efficient)	Vig-aware framing (50.8% needed context) ^[^42]	Medium; important null	Caution against assuming inefficiency is default
NFL under-covers slightly historically	Peer-reviewed paper	Closing line implied	NFL, 1979-2000	>50% win rate, not statistically proven fair-bet rejection	Discussed explicitly ^[^45]	Medium; explicitly inconclusive	Do not treat as executable edge
Wind >13-15mph favors under	Industry blog/dataset	Unclear/likely close-adjacent	CFB, since ~2005, unspecified N	Win rate + ROI claim	Not disclosed	Low [anecdotal1]	Directionally plausible prior only; needs independent test

Execution constraints affecting all of the above

CFBD's own tiered access documentation shows that historical data confidence, call volume, and endpoint access vary by paid tier, meaning the completeness of stored historical lines may itself differ by season and by the developer's own historical pull dates — a real risk of survivorship/coverage bias that must be audited before any line-based backtest. The Odds API's historical odds coverage beginning only in mid-2020 for NCAAAF means that any claim of a "sharp book" leading price discovery, or of cross-book dispersion, can only be independently verified with paid, timestamped multi-book data from mid-2020 forward; pre-2020 sharp-book-vs-recreational-book comparisons are not verifiable with currently identified free/cheap sources, and should be labeled [\[uncertain\]](#) or excluded rather than asserted.^{[^33][^35][^37]}

3. Applicable modeling approaches and prediction targets

Prediction-target comparison

Target	Decision relevance	Valid inputs at decision time	Loss function	Calibration check	Key limitation
Combined final points	Direct total-line comparison	All pregame features	MSE/MAE	Residual-vs-predicted plots	Low RMSE ≠ betting edge if market already near-optimal on same inputs
Separate home/away scores	Enables team-total and joint distribution construction	Same as above, doubled	Joint MSE, correlation-aware	Residual correlation check	Requires modeling score dependence explicitly (Section on score-generating models)
Expected possessions × points/possession	Decomposes total into interpretable structural pieces	Tempo + efficiency features only	Component-wise MSE	Compare implied total to actual	Compounding estimation error across two multiplied components

Target	Decision relevance	Valid inputs at decision time	Loss function	Calibration check	Key limitation
Residual (final total – market total)	Directly tests incremental info beyond market	Market total (opener/current, matched to decision time) + features	MSE on residual	Residual should have mean ~0, no residual autocorrelation with features	Only valid if market-total input is genuinely available at decision time, not leaked from close
Over/under/push probability at available line	Directly maps to bettable decision	Market total/price + features	Log loss/Brier	Reliability diagram, PIT	Probability output must be converted with real price to get EV; probability accuracy alone is not EV
Full predictive distribution / quantiles	Supports variance-aware sizing and multiple lines	All features	CRPS, pinball loss	Coverage tests at multiple quantiles	Harder to fit reliably with limited FBS sample per team-season
Closing total (as separate target)	Diagnostic only — not a bet target	Opener/early features (must not use info only available at close)	MSE	Compare early-model forecast of close vs. actual close	Must never feed into the opener-time betting model as an input; useful only for studying market movement itself
Future line movement / probability of favorable movement	Supports timing decisions (bet now vs. wait)	Only information available at time of forecast	Classification loss	Backtested hit rate on movement direction	High noise-to-signal in single-game movement prediction
Direct EV at current total and price	Final decision variable	Model total/probability + actual live price	N/A (decision rule, not fitted)	EV realized vs. predicted EV over many bets	Requires trustworthy probability calibration, or errors compound into wrong bets

Low RMSE in predicting raw points does not imply betting edge because the sportsbook total is itself a highly informed forecast; a model can have excellent raw accuracy yet contain zero information beyond what the market total already encodes, in which case its residual against the market total is centered at zero with no exploitable signal. This is the central logical distinction the entire report rests on, consistent with the "four evidence levels" framing requested.

Market baselines and calibrated market models

Using the raw market total (opener, current, or closing depending on decision point) as the primary baseline is mandatory, since all comparisons must be benchmarked against it. Calibrating the market total with a simple intercept/slope regression (i.e., checking whether the market is systematically biased high or low in specific total ranges, weeks, conferences, or weather conditions) is a low-cost, high-interpretability first modeling step and is directly motivated by the documented CFB team-totals censoring bias and NFL extreme-total bias findings as plausible (not confirmed) analogues for the combined total. Market-residual forecasting (predicting final total minus market total) is meaningfully different from predicting final points with market total as a feature only insofar as the residual framing forces the modeler to interpret coefficients directly as "edge over market," which is clearer for CLV-oriented development, though mathematically the two are closely related when the market total is included linearly in a full-points model.^{[41][39]}

Linear, regularized, and interpretable models

Ridge/lasso/elastic net regression are well-suited to the FBS sample-size regime (roughly 800-900 FBS games per season, ~130+ teams with substantial year-to-year roster turnover), since regularization controls overfitting given many correlated efficiency features and limited games per team. Hierarchical/mixed-effects and Bayesian partial-pooling models are well-motivated for team-level effects (each team-season has limited games, and partial pooling toward conference or FBS averages directly addresses the shrinkage need identified throughout Section 1). Dynamic/state-space (Kalman-filter-style) models for latent offense/defense/tempo ratings are a natural fit for a longitudinal team-strength framework that updates weekly rather than being refit from scratch each season, directly matching the walk-forward requirement in Section 4; no FBS-totals-specific validated implementation was found in this research, so this remains "applicable but unproven" rather than "demonstrated."

Tree, boosting, and similarity models

Gradient boosting (XGBoost/LightGBM/CatBoost) can handle nonlinear interactions (e.g., pace × efficiency, spread × total) without manual specification and handle missing data natively, useful given inconsistent early-season/sparse-team coverage. Given FBS's roughly 800-900 games/season, tree ensembles risk overfitting to noise unless heavily regularized (shallow trees, strong L1/L2, early stopping on a genuinely chronological validation fold) — with total available seasons of 2012-2025, only a subset of "modern-era" comparable seasons (post-realignment, post-COVID, post-2023-rule-change) may be usable together without introducing regime-shift noise. Quantile/probabilistic boosting (e.g., NGBost, LightGBM quantile objective) is directly applicable to the quantile/distribution targets in Section 3's target table and should be prioritized over plain point-estimate boosting if uncertainty-aware staking is a design goal.

Score-generating, possession, drive, and simulation models

Bivariate/correlated Poisson models are the standard soccer-analytics approach for jointly modeling two dependent team scores rather than assuming independence, and are conceptually transferable to football, but football scoring occurs in discrete increments (3, 6, 7, 8 points) rather than soccer's single-goal counting process, so a naive Poisson-goals analogy is a weaker mechanical fit for football than for soccer — this is a structural caveat, not a disqualifier, since a modified Poisson/negative-binomial process for possessions (not raw points) combined with a separate points-per-possession distribution more faithfully respects football's scoring mechanism. Score dependence in football arises mechanically through shared pace (more plays for one team compressing time for the other), field position/turnover effects, and game-script-driven late-game strategy (leading teams running clock, trailing teams passing/going for it more), all of which argue for explicit joint or copula-based modeling rather than independent team-score assumptions. Overtime rules (including the 2021 shift to alternating 2-point conversion attempts after a second overtime) create nonstandard right-tail scoring behavior that a simple Poisson tail would mis-model; garbage-time-adjusted possession models are consistent with the mechanism evidence discussed in Section 1. No FBS-specific validated backtest of a possession/drive-level Monte Carlo total simulator against the market was found in this research; this entire family is "applicable but unproven" pending direct implementation and walk-forward testing.^{[48][49][50][6][7][5]}

Probability, uncertainty, and calibration methods

Calibration methods (isotonic regression, Platt scaling, reliability diagrams, Brier score, log loss, CRPS, PIT diagnostics) are essential complements to any probability-producing model, but they explicitly do not create new predictive information — they only correct a model's stated confidence to match its empirical accuracy. This distinction matters directly for the "level 3 vs level 4" evidence hierarchy: a perfectly calibrated model with zero information beyond market still produces zero long-run EV after vig.

Neural and representation-learning models

Given FBS's modest annual sample (roughly 800-900 games/season) relative to typical deep-learning data requirements, and substantial year-to-year nonstationarity from coaching changes, transfer portal turnover, and rule changes, complex neural architectures (transformers, GNNs, sequence models) face a high overfitting/instability risk without a correspondingly large and clean dataset. No FBS totals-specific deep-learning backtest was found; this family is "weak fit" for the data-availability constraints stated, though multi-task learning (jointly predicting home score, away score, total, and variance from shared representations) is a reasonable "secondary candidate" if simpler models plateau.

Hybrid and ensemble approaches

Market-plus-machine-learning residual modeling (using the market total as a strong prior/baseline feature, then modeling only the residual with a regularized or boosted model) is the most defensible practical framework given the strength of market-efficiency evidence generally and the specific documented CFB team-totals bias: it inherits the market's substantial informational content while allowing targeted testing of whether specific engineered features (tempo, garbage-time-adjusted efficiency, weather, roster continuity) add incremental signal. This should be a "priority experiment" candidate given it directly operationalizes the central research question.^[^39]

Decision frameworks

Fixed-edge and probability-threshold rules, expected-value/price-aware rules, and fractional Kelly staking are all standard, well-understood, low-engineering-cost frameworks appropriate for a solo developer; full/aggressive Kelly is inappropriate given the estimation uncertainty in a nascent model's true win probability, and fractional Kelly (e.g., quarter-Kelly) or flat/capped-proportional staking are more robust starting points. Reinforcement learning and direct policy learning are explicitly not recommended for this one-shot pregame decision structure: RL's comparative advantage lies in sequential, multi-step decision environments with abundant interaction data, neither of which describes a single per-game total bet decision with an annual sample in the hundreds; this should be treated with the skepticism the query itself calls for.

Model family table

Model family	Best target	Potential edge source	FBS totals evidence	NFL/transfer evidence	Data burden	Main risk	Recommended priority
Market-only calibration	Over/under probability, EV	Correcting known biases (censoring, extreme totals)	Adjacent evidence ^[^39]	Direct NFL evidence ^[^41] ^[^44]	Low	Bias may already be arbitrated away by book	Priority experiment
Regularized linear/GLM on engineered features	Residual vs. market total	Interpretable incremental signal	None direct	None direct	Medium	Feature multicollinearity with market info	Priority experiment
Gradient boosting (quantile/distributional)	Full distribution, quantiles	Nonlinear interactions	None direct	None direct	Medium-high	Overfitting on ~800 games/season	Secondary candidate
Possession/drive simulation	Combined points via structural decomposition	Mechanistic fidelity to football scoring	Mechanism-only ^[^5] ^[^6]	Soccer/general sports analogues only ^[^48] ^[^49]	High	Component error compounding	Secondary candidate
Bayesian hierarchical/state-space team ratings	Latent strength, updated weekly	Proper shrinkage, walk-forward native	None direct	None direct	Medium-high	Complexity vs. benefit unproven	Conditional candidate
Deep learning (MLP/transformer/GNN)	Any	Representation learning	None found	None found	High	Sample size insufficient, unstable	Weak fit
Reinforcement learning / policy learning for staking	Bet/no-bet policy	Sequential optimization	None found	None found	High	Mismatched to one-shot decision structure	Not recommended
Market-plus-ML residual hybrid	Residual/EV	Combines market strength with targeted feature testing	Indirect via market bias evidence ^[^39]	Indirect	Medium	Residual may still be near-zero signal	Priority experiment

4. Validation and leakage pitfalls specific to totals

Time and information leakage

The single highest-risk leakage vector specific to this project is using closing-total fields (or any statistic revised/finalized after kickoff) as an input feature for a strategy intended to fire at the opener or early-week market; this must be structurally prevented in the DuckDB schema by storing separate, clearly timestamped opener/current/close columns and by writing model-input queries that explicitly filter on a `snapshot_time <= decision_time` condition rather than trusting column naming alone. Per-game statistics (like season-to-date EPA) must exclude the game being predicted and, ideally, exclude any statistic that could not have been computed and published before that week's decision time; opponent-adjustment ratings fit on a full season (rather than only games completed before the decision date) are a classic and easy-to-miss leakage source, since a rating system trained on all 15 weeks of a season implicitly "knows" future results when applied to an early-season game in that same season.

Distribution shift and changing FBS environment

The 2023 clock-rule changes are a documented, dated structural break in pace/scoring, and conference realignment (documented broadly in sports media, not independently re-verified here beyond general knowledge) changes the meaning of "conference" as a grouping variable across the 2012-2025 window — any conference-level feature or subgroup analysis must use season-appropriate membership, not present-day labels, exactly as the user's brief specifies. COVID-affected 2020 scheduling (documented broadly as an unusual season with altered schedules and opt-outs, not independently re-sourced here) is a second known distortion window that should be flagged or excluded/tested separately rather than pooled silently with normal seasons. Roster-turnover acceleration from the transfer portal is repeatedly described in current reporting as intensifying year over year, meaning the stationarity assumption behind any team-continuity feature is weaker in 2021-2025 than in 2012-2016, and this nonstationarity should itself be tested (e.g., does a "returning production" feature's predictive value decay over the sample window?) rather than assumed constant.^{[^2][^3][^4][^51][^30][^52][^1]}

Selection bias and repeated research use

The plan must predeclare feature sets, thresholds, and subgroups before looking at chronologically later folds, and must resist the well-documented temptation (illustrated by the anecdotal wind-threshold examples in Section 2) to narrow a subgroup after seeing which cut produces the best win rate; report neighboring thresholds and multi-season stability rather than a single maximum, consistent with the user's own required procedure.

Recommended chronological validation framework

A rolling/expanding-window walk-forward design is appropriate: use an initial training block (e.g., 2012-2018) purely for feature engineering and model-family selection, an inner validation block (e.g., 2019-2021, explicitly including the COVID season as a stress test rather than excluding it silently) for hyperparameter and threshold tuning, and a strictly untouched outer test block (e.g., 2022-2025, encompassing the post-clock-rule-change era) evaluated only once the entire pipeline (features, model, calibration, thresholds, staking) is frozen. Retraining cadence should be weekly within a season (updating team-strength ratings after each week's games) but model architecture/hyperparameters should be frozen for a full season at a time to avoid in-season re-optimization leakage. Week 0/Week 1 and sparse-history teams (new FBS entrants, teams with coaching changes) should be modeled with heavier shrinkage toward preseason priors (recruiting, returning production, prior-year rating regressed to conference mean) rather than excluded outright, but the tradeoff of inclusion vs. exclusion should itself be empirically tested with a robustness comparison, per the user's explicit instruction not to assume a fixed no-bet rule. Given that 2022-2025 data may already have informally influenced typical CFB modeling intuition industry-wide (though not necessarily this specific developer's own prior work), the plan should explicitly state whether any of that window has already been "seen" during feature brainstorming and, if so, treat any strong outer-test result on that specific window with added skepticism and seek a genuinely prospective 2026-forward confirmation period.

Uncertainty estimation

Block bootstrap by week or season (rather than i.i.d. game-level bootstrap) is appropriate because games within a week/season are not independent (shared team-strength estimation error, shared weather patterns, correlated market conditions); this should be used to generate confidence intervals around CLV, win rate, and ROI rather than treating each game as an independent draw. Given the modest annual sample (~800-900 FBS games/season), single-season ROI/CLV point estimates will have wide uncertainty bands, and multi-season aggregation (with per-season breakdowns reported, including worst-season performance) is necessary before drawing confidence in any strategy.

Risk/pitfall table

Risk or pitfall	How it creates false edge	Detection test	Prevention	Residual limitation
Closing line leaked into opener-time model	Model implicitly "knows" future market info	Compare model performance with/without suspect field; test correlation of feature with future close	Explicit snapshot-time filtering in schema/queries	Requires trustworthy raw timestamp data (itself unverified, Section 2)
Full-season opponent-adjusted ratings used mid-season	Rating "sees" future game outcomes	Rebuild ratings using only prior-week data; compare	Strict as-of rating recomputation each week	Computationally heavier; must be automated carefully
Garbage-time-inclusive stats used as skill proxy	Distorts true team ability	Compare filtered vs. unfiltered feature predictive power	Explicit garbage-time definition and filter ^[^5] ^[^6]	Garbage-time definition itself somewhat arbitrary/undocumented consensus
Threshold/subgroup mining (e.g., wind, total range)	Best-looking subgroup mistaken for real edge	Test neighboring thresholds; require multi-season stability	Predeclare thresholds before test fold	Some true effects may be genuinely narrow and get incorrectly penalized
Pooling pre/post 2023 clock-rule seasons without control	Regime shift misread as team/feature signal	Segment analysis by rule era; include rule-era dummy	Explicit regime flag feature; consider separate models	Reduces effective sample size if segmented

Risk or pitfall	How it creates false edge	Detection test	Prevention	Residual limitation
Repeated 2022-2025 holdout reuse across many project iterations	Overfitting to a "final" test set via researcher's own iteration	Track how many times a given holdout has been evaluated	Reserve a genuinely fresh prospective period (2026+)	Solo developer must self-audit; no external enforcement
Retrospective best-price selection (using best historical book price, not what was executable)	Inflated backtest ROI vs. realistic execution	Compare single-book vs. best-of-book backtest results	Use single consistent book or documented consensus methodology	Even documented consensus lines may not reflect actually accessible account limits

CLV and outcome evaluation

Point CLV is defined directionally as closing total minus bet total for an over position and bet total minus closing total for an under position, matching the user's own specification; positive point CLV does not prove profit because it says nothing about win probability or price paid, and realized short-run profit does not prove genuine model edge because of high game-to-game variance relative to a plausible true edge size. Negative CLV can occur even with a genuinely valid forecast if execution timing is poor (e.g., betting into a stale screen right before a major line move) or if the book chosen is a slow-moving recreational book whose "close" lags sharp consensus — CLV should therefore be benchmarked against a clearly defined, consistent close source (ideally sharp-book or consensus close) rather than an arbitrary single book's potentially stale closing number.

5. Free or cheap data sources beyond CFBD

Source	URL	Fields	Historical coverage	Timestamp/as-of quality	Cost/access	Join method
CollegeFootballData.com API	https://api.collegefootballdata.com ^[^32]	Games, teams, players, recruiting, rankings, lines, PBP-derived stats	Documented "Data availability" page exists but exact per-endpoint year coverage not independently confirmed here	Not confirmed at field level for lines endpoint in this research	Free tier 1,000 calls/month; paid Patreon tiers for more calls/GraphQL ^[^33]	Game ID, 1 season/week
The Odds API historical odds	https://the-odds-api.com/historical-odds-data/ ^[^34]	Moneyline, spread, total, multi-book prices with timestamps	NCAAF featured markets from mid-2020 only ^[^35] ^[^37]	High — explicit timestamped historical snapshots	Paid tiers (\$30-59+/month cited) ^[^53] ; historical query costs 10 credits/region/market ^[^34] ^[^36]	Team names/dates matching CFBD game (manual mapping needed)
Visual Crossing Weather API (historical observed)	https://www.visualcrossing.com/weather-data/ ^[^15]	Temp, wind, precip, humidity, conditions by lat/long/date	50+ years of history claimed ^[^16]	Good day/hour granularity for observed data	Free: 1,000 records/day / 30,000/month ^[^16] ^[^18] ; paid tiers for volume	Stadium location + game date
Visual Crossing Historical Forecast	https://www.visualcrossing.com/weather-data-pricing/ ^[^16]	Forecast-as-issued snapshots from historical dates	Available as distinct paid product	High — designed exactly for this use case	\$165/month cited tier ^[^16]	Stadium location + forecast date
Meteostat	https://dev.meteostat.net/data ^[^17]	Station-based historical observed weather (hourly/daily)	Open historical archive, station-dependent	Good for observed data; not a forecast product	Free/open access ^[^17] ^[^19] ; also available via RapidAPI wrapper ^[^14]	Nearest station to stadium lat/long

Source	URL	Fields	Historical coverage	Timestamp/as-of quality	Cost/access	Join metth
ESPN returning production / returning snap %	https://www.espn.com/college-football/... (annual article)[^23]; CBS Sports[^24]	Team-level preseason production/snap continuity percentiles	Published annually preseason; long-running Bill Connelly methodology	Preseason publication date typically confirmable via article date	Free (public articles)	Team name/season match to team IDs
Recruiting ratings (247Sports Composite, etc.)	Referenced via CFBD recruiting endpoint and third-party sources[^25][^26]	Class/player star ratings, composite scores	Long historical run, exact CFBD coverage not independently confirmed here	Recruiting class dates are well-defined (signing day)	Free via CFBD if endpoint covers it; otherwise scrape	Team/class join
Transfer portal trackers	e.g., CBS Sports transfer portal coverage[^52], On3/247 portal trackers (not independently verified here)	Player entry/exit dates, destination	Short history (portal itself is recent, roughly 2018+ with major growth 2021+)	Moderate; dates generally reported near real-time but not centrally archived with consistent schema	Free (public reporting), no dedicated API confirmed	Player/team matching, manual/sc
FBS injury/depth chart archives	None identified with confirmed historical as-of coverage	N/A	Not found	Not found	N/A	N/A

Prospective collection plan (where historical data does not exist)

For injuries/depth charts, transfer-portal-driven roster shocks, and archived weather forecasts, the recommended path is prospective collection beginning immediately: poll a fixed set of public depth-chart/injury-report sources (e.g., official team releases, conference injury report pages where they exist) at a consistent daily or twice-weekly cadence during the season, store each poll with an explicit UTC timestamp, source URL, and raw text/structured snapshot in a DuckDB table keyed by team_id/date/source; separately poll a chosen weather-forecast API (ideally the same provider used for any future paid historical-forecast subscription, for internal consistency) at a fixed lead time before kickoff (e.g., 5 days out and again 24 hours out) and store both snapshots with issue-time and target-game-time fields. Retention should be indefinite (storage cost is trivial relative to modeling value), and these prospective records should be excluded from any claim of "historical validation" until enough seasons accumulate (a minimum of 2-3 full seasons is a reasonable, though unproven, rule of thumb before using such prospective-only features in a walk-forward test) [uncertain – no external benchmark found for the minimum].

6. Ranked list of five experiments

Rank	Experiment	Hypothesis	Target	Decision time	Baselines	Primary metric	Data burden	Main ris
1	Line-field audit and market-only calibration	CFBD line fields are trustworthy and the raw market total shows exploitable calibration bias in specific subgroups (e.g., extreme totals, censoring-adjacent team-total construction)	Over/under probability vs. market total; calibration slope/intercept	Opener and current/day-of-game, compared separately	Raw market total unadjusted	Calibration slope/intercept, Brier score, CLV	Low (uses existing CFBD data only)	Line field opener/c distinctic downstre
2	Garbage-time-adjusted	Garbage-time-filtered EPA/success-rate/pace features	Combined final points	Entering-game, using	Market total alone;	RMSE reduction + residual-vs-	Medium (requires PBP processing +	Garbage-itself is

Rank	Experiment	Hypothesis	Target	Decision time	Baselines	Primary metric	Data burden	Main risk
	tempo/efficiency ablation	predict raw points better than unfiltered, and this improvement is not simply a market-known effect	(residual vs. market)	only data through prior week	unfiltered-feature model	market-total independence test	garbage-time definition)	subjective consensus
3	Possession/points-per-possession structural simulator	Decomposing the total into possessions × points-per-possession with explicit opponent adjustment produces a full predictive distribution with better calibration than a single point estimate, even if mean-level accuracy is similar to a linear model	Full distribution / quantiles of total	Entering-game	Market-implied distribution (from total + price); Gaussian residual approximation around linear-model point forecast	CRPS, PIT coverage	High (drive-level modeling, overtime/garbage-time handling)	Component tail miss
4	Weather-mechanism magnitude study (observed-weather-only, non-betting)	Wind/temperature/precipitation, measured via cheap observed-weather data, correlates with lower total scoring at specific thresholds, independent of any betting claim	Combined final points vs. observed weather bins	Postgame observed weather used strictly as a mechanism study, NOT as a pregame feature in this experiment	None (this is a diagnostic, not a betting test)	Correlation/effect size with confidence intervals by bin	Low (free Meteostat/Visual Crossing observed data)	Must not a forecast pregame accident; framing
5	Opener-vs-later-market timing comparison	Waiting until a later pregame decision point (e.g., 24 hours pre-kickoff) versus betting at the opener produces different average CLV and different realized ROI, for an otherwise identical model/threshold	CLV (points) and ROI at two decision times using the same underlying forecast	Compare opener snapshot vs. day-of-game snapshot explicitly	Same model at both timing points; no separate baseline needed beyond internal comparison	CLV distribution difference, ROI difference, bet-count difference	Medium (requires reliable multiple-snapshot line data, ideally from mid-2020 The Odds API window or audited CFBD snapshots)	Field-timing (again depends on experimental result)

Experiments 1 and 5 both involve market-timing questions but test different things: Experiment 1 asks whether the market total itself is biased at a single point in time (a calibration question), while Experiment 5 asks whether the choice of when to transact changes realized outcomes given a fixed model (an execution-timing question) — both are necessary and non-redundant.

First experiment to run

Line-field audit and market-only calibration.

This ranks first because every other experiment in this report, and effectively every future modeling decision in this project, depends on knowing whether the CFBD-stored line fields actually represent what they claim (specific sportsbook, correct opener/closing designation, reliable timestamp) — without this, "opener-time" and "closing" claims elsewhere in the pipeline cannot be trusted, and CLV cannot be computed correctly. It resolves the most basic open question raised throughout this report: is the foundational market data trustworthy enough to build anything else on top of it? Data is sufficient because it uses only the data already stored (2012-2025 CFBD line fields), requiring no new acquisition, and cross-validation against The Odds API's mid-2020-forward, timestamped multi-book data provides an independent check for at least the post-2020 portion of the sample. If the hypothesis fails (i.e., fields are unreliable, mislabeled, or systematically stale), the result remains highly useful because it prevents wasted engineering effort on downstream models built on faulty market features and redirects effort toward either sourcing better line data or explicitly restricting the usable historical window. The exact first deliverable is a DuckDB audit report comparing CFBD-stored "opening" and "closing" total fields against The Odds API's historical multi-book snapshots for the overlapping mid-2020-2025 window, quantifying discrepancy rates, missingness, and any systematic staleness, plus a documented decision on which seasons/fields are usable going forward. The freeze-before-test rule: the audit methodology (which discrepancy threshold counts as "unreliable," which comparison window is used) must be fully specified and written down before running the comparison, and no field-selection decision should be revised after seeing which choice makes the data "look better."

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Final quality check notes

All numeric claims above are attributed to their source and labeled [anecdotal] or [uncertain] where methodology, sample size, line timing, or out-of-sample structure was not confirmed. NFL findings are explicitly separated from CFB findings throughout and never presented as confirmed CFB effects. No claim of a strategy beating the market is made without the caveat that Arscott's team-totals result is the only Level-4 (post-vig profit) evidence found, and it applies to team totals rather than confirmed for the standard combined-game total market. CFBD line-field trustworthiness (sportsbook identity, opener/close designation, timestamp) could not be independently verified from the documentation pages reviewed and is flagged as the first required diagnostic rather than assumed.

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